SIMATS ENGINEERING
Engineer to Excel
SAVEETHA TRANSDISCIPLINARY ANNUAL RESEARCH SUMMIT
STAR SUMMIT 2024
SIMATS ENGINEERING
TECH STAR SUMMIT 2024

Name: Mr. Bharathwaj.K.S
Register Number: 192124033
Guided by Dr.S.Kalaiarasi

Enhancing Prediction Accuracy in Civil Case Judgements through the Implementation of Extreme Gradient Boosting and Comparison with Random Forest Algorithm.

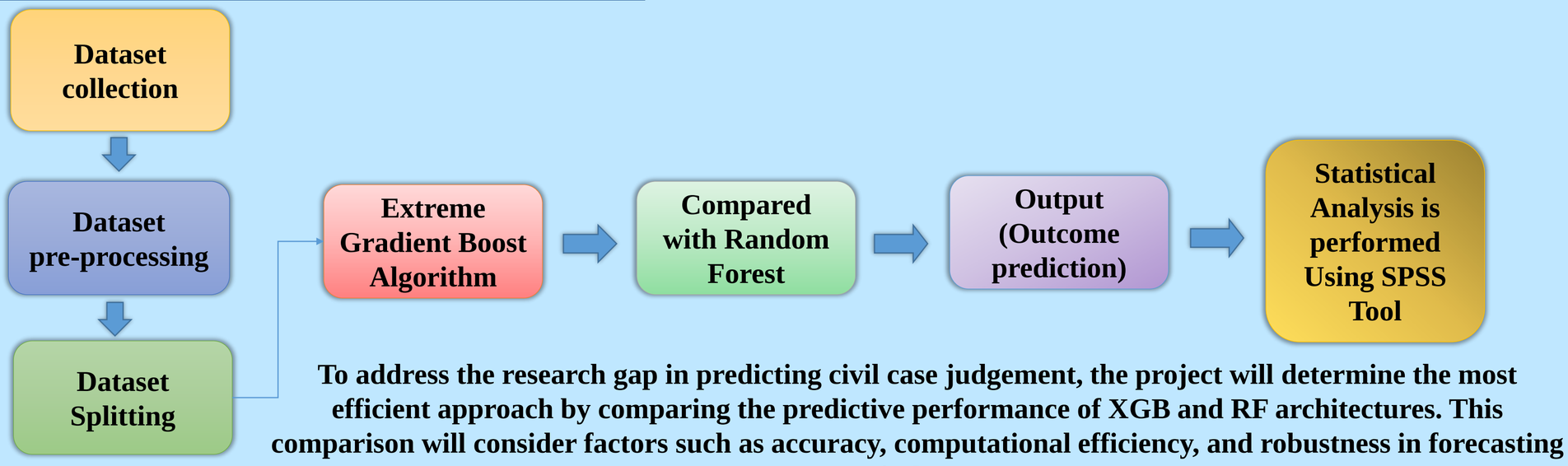
INTRODUCTION

- The aim of this project is to enhance the accuracy of predicting civil case judgments by implementing the Extreme Gradient Boosting (XGB) algorithm and comparing its performance with the Random Forest (RF) algorithm.
- Civil case proceedings often encounter prolonged delays, leading to inefficiencies and resource wastage. These delays escalate legal expenses, erode public confidence in the legal system, and hinder access to justice. Therefore, there is a pressing need to develop efficient predictive models to streamline the judgment process and mitigate the associated challenges.
- The application of this project extends to various sectors and industries where civil case judgments play a crucial role. By accurately predicting civil case outcomes, this project can facilitate timely resolutions, reduce legal costs, and enhance access to justice for all stakeholders involved.
- The Extreme Gradient Boosting (XGB) algorithm is chosen for its ability to handle complex data patterns, high scalability, and robustness in predicting outcomes. It combines multiple weak learners to improve accuracy and generalization. Additionally, XGB offers interpretability and simplicity, making it suitable for practical applications in legal prediction tasks.
- The dataset utilized in this project is sourced from Kaggle, containing a comprehensive collection of civil case data. This dataset encompasses various attributes relevant to civil case judgments, including parties involved, legal documents support, lawyer experience, and decision types.

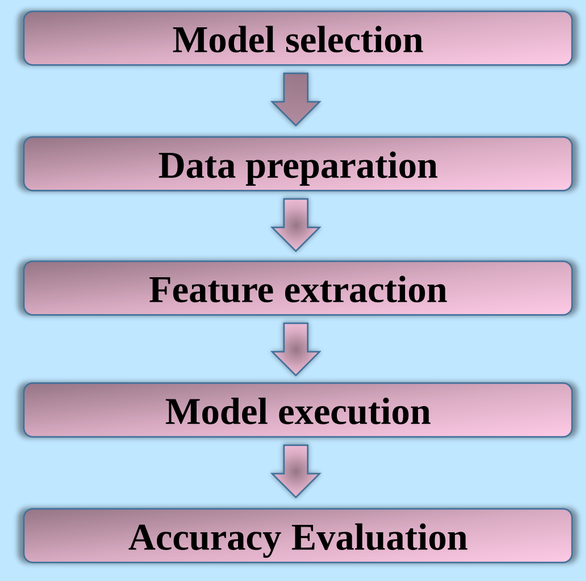


Civil cases in courts of India

MATERIALS AND METHODS



To address the research gap in predicting civil case judgement, the project will determine the most efficient approach by comparing the predictive performance of XGB and RF architectures. This comparison will consider factors such as accuracy, computational efficiency, and robustness in forecasting court judgements.

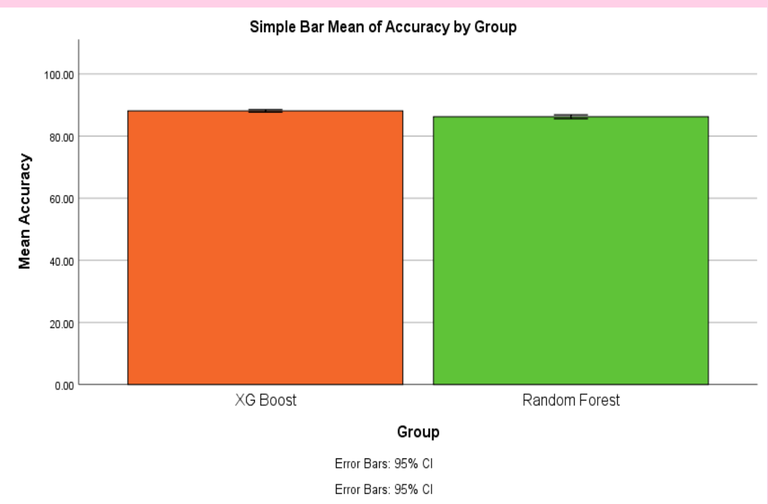


RESULTS

- To conduct SPSS analysis, gather a sample size of 10 accuracies for each algorithm. Compare them using a paired sample t-test in SPSS. Calculate mean accuracy, standard deviation, and plot them using a graph.
- The obtained significance value, 0.001, indicates a statistically significant difference between the mean accuracies of the two algorithms in predicting water inundation frequency. This suggests that one algorithm significantly outperforms the other, emphasizing the preference for choosing the algorithm with higher accuracy for practical implementation or further development in water inundation frequency analysis.

	Group	N	Mean	Std. Deviation	Std. Error Mean
Accuracy	XG Boost	10	88.4901	0.54099	0.17142
	RF	10	86.2201	0.84362	0.26687

		Levene's Test for Equality of Variances		T-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
Accuracy	Equal variances assumed	2.149	0.162	5.899	18	0.001	1.87850	0.31617	1.20145	2.53625
	Equal variances not assumed			5.999	15.130	0.001	1.87650	0.33617	2.83120	2.54480



XG Boost and RF

DISCUSSION AND CONCLUSION

- The mean accuracy achieved for the Extreme Gradient Boosting (XGB) model is 88.4901 with a standard deviation of 0.54099, while for the Random Forest (RF) model, it is 86.2201 with a standard deviation of 0.84362.
- Looking ahead, there is potential for further advancements in the field by exploring ensemble learning techniques, such as stacking, which combine the strengths of multiple algorithms to improve predictive performance.
- The accuracy of predictive models may be influenced by factors such as the quality and quantity of available data, as well as the specific characteristics of legal proceedings. Moreover, the computational resources required for training and optimizing machine learning models may pose constraints on scalability and efficiency.
- In conclusion, the findings of this study underscore the superiority of Extreme Gradient Boosting (XGB) over Random Forest (RF) in predicting civil case judgments with its higher accuracy and robustness.

BIBLIOGRAPHY

- Abur, D., Hillman, R.E. and Stepp, C.E. (2023) ‘Auditory-Motor Function Pre- and Post-Therapy in Hyper functional Voice Disorders: A Case Series’, Journal of voice: official journal of the Voice Foundation .
- Gonzalez-Hernandez, L.M. and San José, J. (2023) ‘Professional responsibility in obstetric-gynecological nursing: Update of the case law of the High Court of Justice of Andalusia (Spain)’, PloS one, 18(9), p. e0291669.
- Barnett, K.B. et al. (2023) ‘Characteristics of substance-addicted mothers that predict graduation from the family treatment court’, Journal of nursing scholarship: an official publication of Sigma Theta Tau International Honor Society of Nursing / Sigma Theta Tau, 55(3), pp. 637–645.
- Chen, J. et al. (2023) ‘Epidemiological characteristics and dynamic transmissions of COVID-19 pandemics in Chinese mainland: A trajectory clustering perspective analysis’, Epidemics, 45, p. 100719.

SIMATS ENGINEERING
Engineer to Excel
TECH STAR SUMMIT 2024
Name: Mr. Bharathwaj.K.S
Register Number: 192124033
Guided by Dr.S.Kalaiarasi

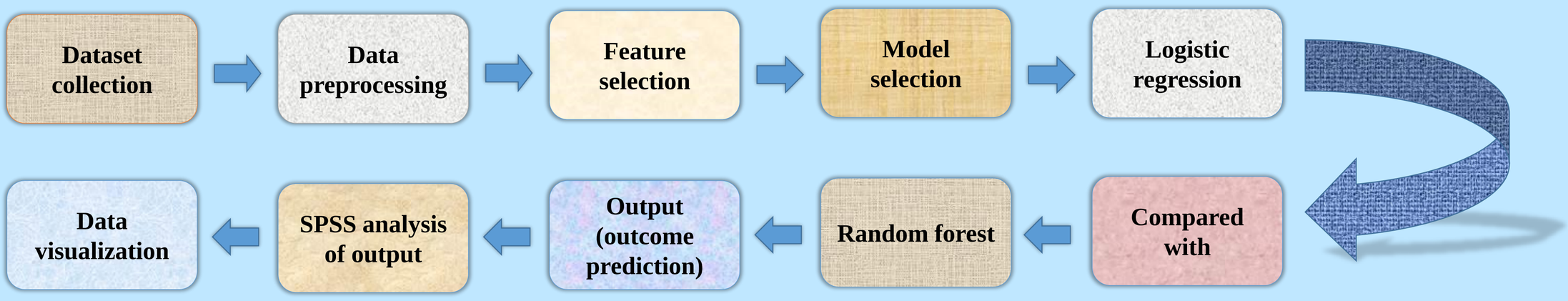
Improved Accuracy for Court Case judgements Forecasting with Logistic Regression Model versus Random Forest Algorithm.

INTRODUCTION

- The aim of this project is to refine the accuracy of civil case judgment predictions by employing the Logistic Regression algorithm and contrasting its effectiveness with the Random Forest algorithm.
- The significance of this project stems from the persistent challenges faced in civil case proceedings, characterized by prolonged delays and resource inefficiencies. These delays not only inflate legal expenses but also undermine public trust in the legal system and hinder timely access to justice.
- The application of this project transcends the legal domain, with potential implications across various sectors reliant on efficient and accurate decision-making processes. By leveraging advanced machine learning techniques, such as Logistic Regression, this project aims to streamline civil case judgments, thereby reducing delays and optimizing resource allocation.
- Logistic Regression is selected for its interpretability, simplicity, and adaptability to diverse datasets. It excels in modeling binary outcomes and offers insights into the influence of input variables on the predicted outcomes. Additionally, Logistic Regression can handle nonlinear relationships between variables, making it well-suited for complex legal prediction tasks.
- The dataset utilized in this project is sourced from Kaggle, encompassing a diverse range of attributes pertinent to civil case judgments. These attributes include the identities of involved parties, legal documentation support, lawyer expertise, and decision types.


Civil cases in High courts of India

MATERIALS AND METHODS



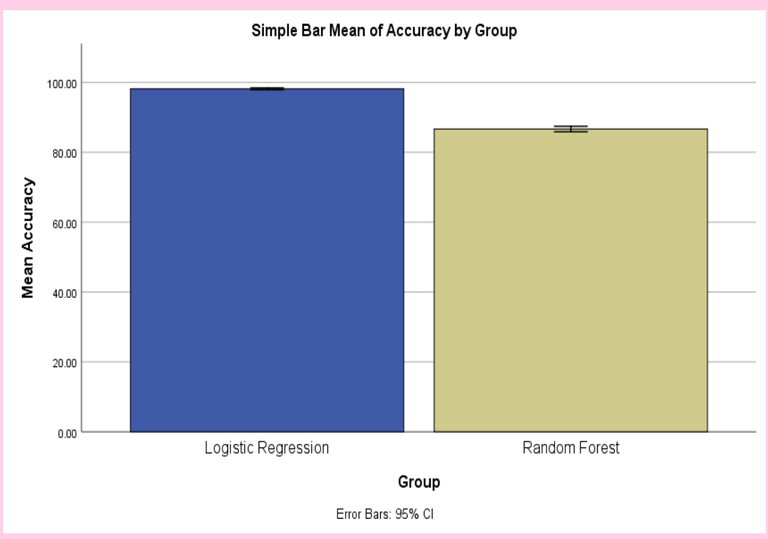
To address the research gap in predicting civil case judgement, the project will determine the most efficient approach by comparing the predictive performance of LR and RF architectures. This comparison will consider factors such as accuracy, computational efficiency, and robustness in forecasting court judgements.

RESULTS

- To conduct SPSS analysis, gather a sample size of 10 accuracies for each algorithm. Compare them using a paired sample t-test in SPSS. Calculate mean accuracy, standard deviation, and plot them using a graph.
- The obtained significance value, 0.001, indicates a statistically significant difference between the mean accuracies of the two algorithms in predicting water inundation frequency. This suggests that one algorithm significantly outperforms the other, emphasizing the preference for choosing the algorithm with higher accuracy for practical implementation or further development in water inundation frequency analysis.

	Group	N	Mean	Std. Deviation	Std. Error Mean
Accuracy	LSTM	10	97.8160	0.3599	0.11321
	RBFNN	10	86.2020	1.09419	0.34601

		Levene's Test for Equality of Variances		T-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
Accuracy	Equal variance assumed	7.011	0.16	31.599	18	0.001	11.50400	0.36320	10.73913	12.52689
	Equal variance not assumed			31.564	10.905	0.001	11.807000	0.36740	10.45063	12.52040



LR and RF

DISCUSSION AND CONCLUSION

- The mean accuracy obtained for the LR model is 97.8160 with a standard deviation of 0.3599, whereas for the RF model, it is 88.4810 with a standard deviation of 2.99839.
- For future advancement exploring cloud-based solutions for real-time prediction systems in legal proceedings, leveraging scalable infrastructure and distributed computing to provide instantaneous insights and support decision-making processes.
- Despite its potential, the accuracy of predictive models may be constrained by the availability and quality of court case data, while training deep learning models like LR and RF may require substantial computational resources.
- In conclusion, that the Logistic Regression attains the high accuracy when comparing with the Random Forest in legal prediction system in civil court cases due to it's ability in modeling binary outcomes and adaptability to diverse datasets.

BIBLIOGRAPHY

- Liao, Boyu, Yen-Chiang Chang, and Mehran Idris Khan. 2023. "The Impact of the Monetary Gold Principle on the Determination of Jurisdiction in the Mixed Maritime Disputes." Science Progress 106 (2): 368504231179814.
- Mersky, Joshua P., Chienti Plummer Lee, Xiyao Liu, and Colleen E. Janczewski. 2023. "Impact of a Family Treatment Court on Child Permanency and Safety." Child Abuse & Neglect 146 (December): 106512.
- Nguyen, David-Dan, Massine Fellouah, Anna-Lisa V. Nguyen, David-Christian Kazu, Isabel Baltzan, Muhieddine Labban, Shubha De, et al. 2023. "Litigation Associated with 5-Alpha-Reductase-Inhibitor Use: A Canadian Legal Database Review." The Canadian Journal of Urology 30 (3): 11546–50.
- Shiely, Frances, Danielle Murphy, and Seán R. Millar. 2023. "Clinical Research Nurse Predictions of Trial Failure, Recruitment and Retention: A Case for Their Early Inclusion in Trial Design." Trials 24 (1): 458.

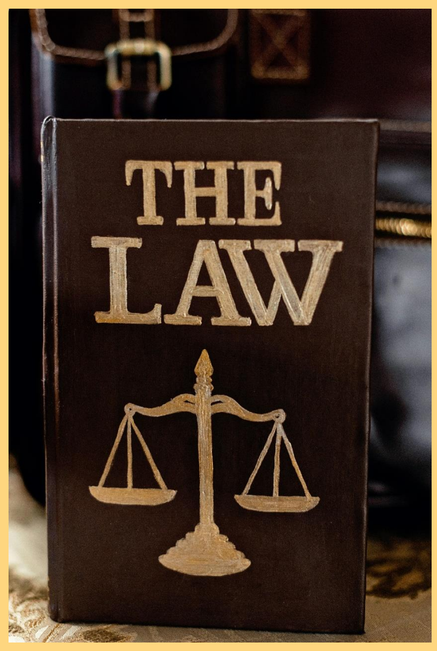
SIMATS ENGINEERING
Engineer to Excel
SAVEETHA TRANSDISCIPLINARY ANNUAL RESEARCH SUMMIT
STAR SUMMIT 2024
SIMATS ENGINEERING
TECH STAR SUMMIT 2024

Name: Mr. Bharathwaj.K.S
Register Number: 192124033
Guided by Dr.S.Kalaiarasi

A Comparative Investigation of Prediction Accuracy in Civil Case Verdicts: Decision Tree Model vs. Random Forest Algorithm to predict the case judgment using a decision tree.

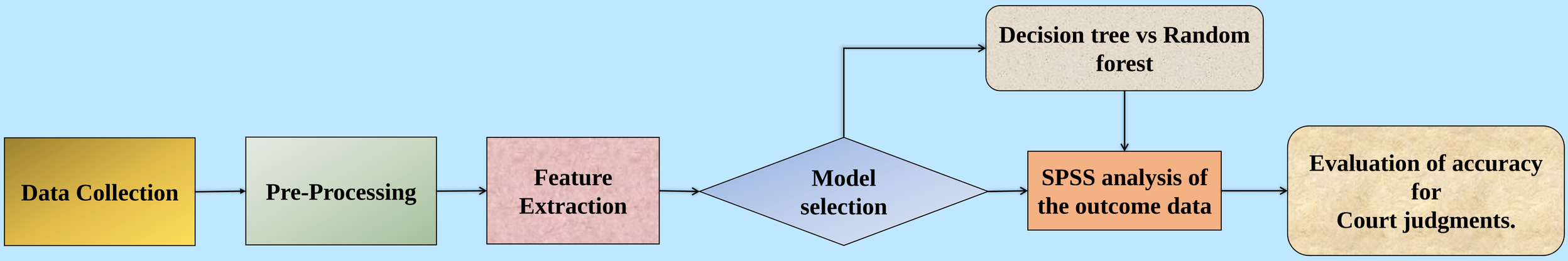
INTRODUCTION

- The primary aim of this project is to enhance the precision of civil case judgment predictions by implementing the Decision Tree algorithm and conducting a comparative analysis with the Random Forest algorithm, which is currently employed.
- In contemporary legal landscapes, civil case proceedings often grapple with prolonged timelines, leading to inefficiencies and resource depletion. These delays escalate legal expenses, erode public trust in the legal system, and impede the timely resolution of disputes. Therefore, it is imperative to develop advanced predictive models to streamline judgment processes and mitigate these challenges.
- The application of this project spans various sectors where accurate judgment predictions are essential. By leveraging the Decision Tree algorithm, this project seeks to expedite resolutions, mitigate legal costs, and facilitate broader access to justice.
- The Decision Tree algorithm is chosen for its simplicity, interpretability, and capability to handle both numerical and categorical data. It excels in identifying complex decision boundaries and offers insights into the decision-making process. Additionally, Decision Trees are robust against overfitting, making them well-suited for legal prediction tasks.
- The dataset utilized in this study is obtained from Kaggle, encompassing a comprehensive array of civil case data. This dataset includes diverse attributes relevant to civil case judgments, such as parties involved, legal documents support, lawyer experience, and decision types.



Civil cases in High courts of India

MATERIALS AND METHODS



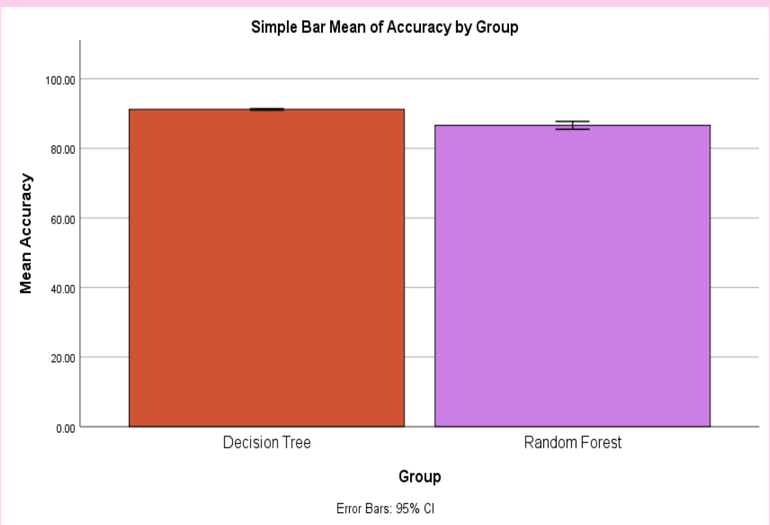
To address the research gap in predicting civil case judgement, the project will determine the most efficient approach by comparing the predictive performance of DT and RB architectures. This comparison will consider factors such as accuracy, computational efficiency, and robustness in forecasting court judgements.

RESULTS

- To conduct SPSS analysis, gather a sample size of 10 accuracies for each algorithm. Compare them using a paired sample t-test in SPSS. Calculate mean accuracy, standard deviation, and plot them using a graph.
- The obtained significance value, 0.001, indicates a statistically significant difference between the mean accuracies of the two algorithms in predicting water inundation frequency. This suggests that one algorithm significantly outperforms the other, emphasizing the preference for choosing the algorithm with higher accuracy for practical implementation or further development in water inundation frequency analysis.

	Group	N	Mean	Std. Deviation	Std. Error Mean
Accuracy	Decision Tree	10	91.1990	0.32209	0.10301
	Random forest	10	86.5740	1.51419	0.49958

		Levene's Test for Equality of Variances		T-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
Accuracy	Equal variance assumed	15.032	0.001	9.021	18	0.001	4.65201	0.53402	3.53521	5.64753
	Equal variance not assumed			9.021	9.027	0.001	4.60500	0.56620	3.46463	5.56434



DT and RF

DISCUSSION AND CONCLUSION

- The mean accuracy achieved for the Decision tree (XGB) model is 91.1990 with a standard deviation of 0.32209, while for the Random Forest (RF) model, it is 86.5740 with a standard deviation of 1.51419.
- Future research could explore the integration of advanced optimization techniques, cloud-based solutions for real-time and feature engineering methods to enhance the performance of prediction models like DT and RF.
- The limitations include the availability and quality of court case data, the need for continuous model refinement, and the computational resources required for training and deployment. Additionally, the interpretability of complex models like DT may pose challenges in legal settings where transparency and accountability are paramount.
- In conclusion, the results demonstrate that the Decision Tree (DT) model outperforms the Random Forest (RF) model in predicting civil case judgments, achieving higher accuracy and demonstrating greater robustness.

BIBLIOGRAPHY

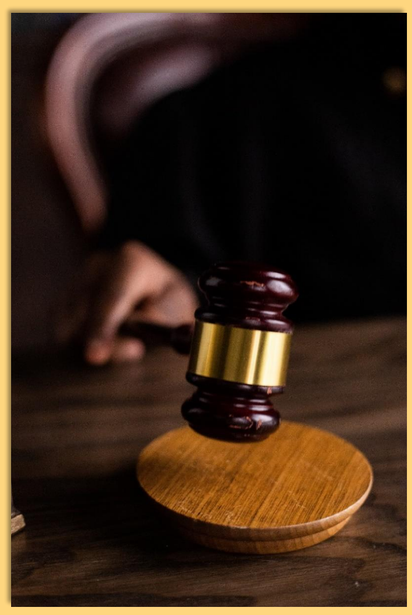
- Hill, E.L. et al. (2023) 'Risk factors associated with post-acute sequelae of SARS-CoV-2: an N3C and NIH RECOVER study', BMC public health, 23(1), p. 2103.
- Hogg, H.D.J. et al. (2023) 'Safety and efficacy of an artificial intelligence-enabled decision tool for treatment decisions in neovascular age-related macular degeneration and an exploration of clinical pathway integration and implementation: protocol for a multi-methods validation study', BMJ open, 13(2), p. e069443.
- Miller, H.E. et al. (2023) 'Abortion Bans and Resource Utilization for Congenital Heart Disease: A Decision Analysis', Obstetrics and gynecology, 142(3), pp. 652–659.
- Zalikha, A.K. et al. (2023) 'Improved performance of machine learning models in predicting length of stay, discharge disposition, and inpatient mortality after total knee arthroplasty using patient-specific variables', Arthroplasty (London, England), 5(1), p. 31.

SIMATS ENGINEERING
Engineer to Excel
TECH STAR SUMMIT 2024
Name: Mr. Bharathwaj.K.S
Register Number: 192124033
Guided by Dr.S.Kalaiarasi

Comparative Analysis of Civil Case Verdict Prediction: Gaussian Naive Bayes classifier Algorithm vs. Random Forest Model for Enhanced Accuracy.

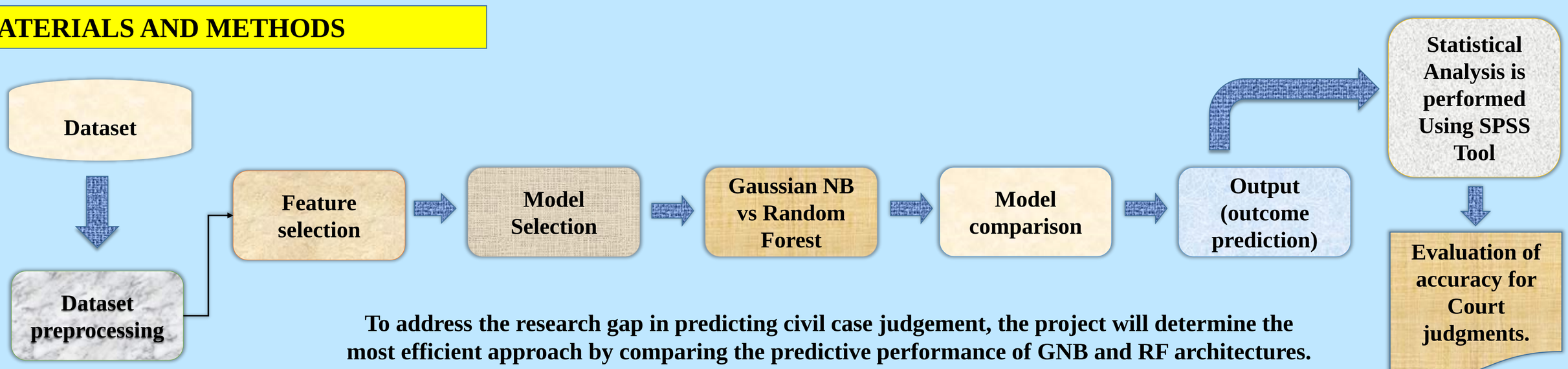
INTRODUCTION

- The primary objective of this project is to refine the accuracy of civil case judgment predictions through the implementation of the Gaussian Naive Bayes (GNB) algorithm and conducting a comparative assessment against the Random Forest (RF) algorithm.
- In today's legal landscape, civil case proceedings often encounter prolonged delays, resulting in inefficiencies and resource depletion. These delays amplify legal expenses, erode public confidence in the legal system, and hinder timely dispute resolution. Hence, there exists a critical need to develop advanced predictive models to expedite judgment processes and alleviate these challenges.
- The scope of application for this project extends beyond the legal sector, encompassing diverse industries where accurate judgment predictions are imperative. By leveraging the Gaussian Naive Bayes algorithm, this project endeavors to expedite resolutions, mitigate legal costs, and foster equitable access to justice.
- The Gaussian Naive Bayes algorithm is selected for its inherent simplicity, interpretability, and effectiveness in handling both numerical and categorical data. It excels in identifying underlying patterns and relationships within data, offering valuable insights for decision-making.
- The dataset utilized in this study is sourced from Kaggle, featuring a comprehensive array of civil case data. This dataset encompasses an extensive range of attributes pertinent to civil case judgments, including parties involved, legal documents support, lawyer experience, and decision types.



Civil cases in High courts of India

MATERIALS AND METHODS



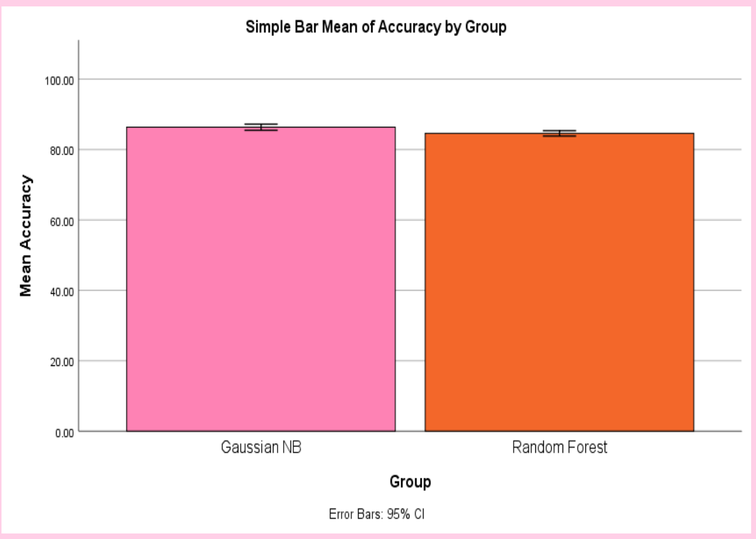
To address the research gap in predicting civil case judgement, the project will determine the most efficient approach by comparing the predictive performance of GNB and RF architectures. This comparison will consider factors such as accuracy, computational efficiency, and robustness in forecasting court judgements.

RESULTS

- To conduct SPSS analysis, gather a sample size of 10 accuracies for each algorithm. Compare them using a paired sample t-test in SPSS. Calculate mean accuracy, standard deviation, and plot them using a graph.
- The obtained significance value, 0.001, indicates a statistically significant difference between the mean accuracies of the two algorithms in predicting water inundation frequency. This suggests that one algorithm significantly outperforms the other, emphasizing the preference for choosing the algorithm with higher accuracy for practical implementation or further development in water inundation frequency analysis.

	Group	N	Mean	Std. Deviation	Std. Error Mean
Accuracy	Gaussian NB	10	88.3850	0.90542	0.28412
	Random forest	10	86.1079	0.92451	0.29420

		Levene's Test for Equality of Variances		T-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
Accuracy	Equal variances assumed	0.124	0.725	0.4204	18	0.001	1.72201	0.40202	2.21250	0.88520
	Equal variances not assumed			.4032	17.995	0.001	1.22032	0.40326	2.65201	0.86102



GNB and RF

DISCUSSION AND CONCLUSION

- The mean accuracy achieved for the Extreme Gradient Boosting (XGB) algorithm is 88.3850, with a standard deviation of 0.90542, while for the Random Forest (RF) algorithm, it is 86.1079, with a standard deviation of 0.92451.
- Looking ahead, there is potential for further advancements in predictive modeling for legal proceedings, particularly in refining the implementation of Extreme Gradient Boosting to handle larger datasets and complex decision-making scenarios. Additionally, integrating XGB with cloud-based solutions could facilitate real-time predictions and enhance the accessibility and efficiency of legal processes.
- The accuracy of predictive models like XGB may be influenced by factors such as data quality, feature selection, and model tuning. Moreover, while XGB exhibits scalability and robustness, it may require careful parameter optimization and computational resources for optimal performance.
- In conclusion, the findings indicate that the Extreme Gradient Boosting algorithm outperforms Random Forest in predicting civil case judgments, as evidenced by its higher accuracy and lower standard deviation due to the adaptability and scalability.

BIBLIOGRAPHY

- Merianos, Ashley L., E. Melinda Mahabee-Gittens, Timothy M. Stone, Roman A. Jandarov, Lanqing Wang, Deepak Bhandari, Benjamin C. Blount, and Georg E. Matt. 2023.
- Partington, Scott, Shaun Nichols, and Tamar Kushnir. 2023. “Rational Learners and Parochial Norms.” Cognition 233 (April): 105366.
- Sabry, Fouad. 2023. Naive Bayes Classifier: Fundamentals and Applications. One Billion Knowledgeable.
- Fang, Kwo-Ting, and Ching-Hsiang Ping. 2022. “Using Machine Learning to Explore the Crucial Factors of Assistive Technology Assessments: Cases of Wheelchairs.” Healthcare (Basel, Switzerland) 10 (11).
- Tang, Yan-Min, Yan-Hong Wang, Xin-Yu Feng, Qiao-Sha Zou, Qing Wang, Jing Ding, Richard Chuan-Jin Shi, and Xin Wang. 2022. “Diagnostic Value of a Vision-Based Intelligent Gait Analyzer in Screening for Gait Abnormalities.” Gait & Posture 91 (January): 205–11